Epistemic Navigation in Open Worlds

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Abstract

Scientific navigation transforms can preserve a local plan while losing the obligations needed for task-global scientific closure. We give a closure-carrying navigation semantics in which a donor-valid transformation carries closure only through an explicit obligation witness. Over uninterpreted sorts, identity is a unit, composition is associative, and a composite carries closure when both legs carry it and their intermediate obligation contracts match or have a registered bridge. The side conditions are explicit: reflexivity supports the unit law, extensionality supports equational forms, and obligation-totality composition requires containment of demanded obligations. A finite five-family interpretation reproduces 25 composition successes and 25 bridge-mismatch countermodels. A negative result is retained: the rule is provably incomplete against its own obligation semantics because it can refuse extensionally equivalent demands when no bridge is registered. The calculus is therefore sound and fail-closed, not exact. It does not claim a new donor navigation mechanism, external validation, or deployed-agent superiority.

Overview and result

Scientific navigation already has strong donor mechanisms: sound planning abstraction and refinement, counterexample-guided refinement, bidirectional transformation and migration, world-model replanning, and terminal-commitment frameworks. This paper treats these as reusable navigation transforms and asks how task-global scientific closure survives them. The answer is a closure-carrying navigation semantics in which a donor-valid transformation carries closure only through an explicit obligation witness, and in which heterogeneous transforms compose only when the intermediate obligation contract is exactly bound or explicitly bridged.

That composition law is now a **theorem rather than an enumeration**. Over uninterpreted sorts of transformations, contracts, closure coordinates and obligations, with the donor-native validity predicate left uninterpreted, identity is a unit, composition is associative, and a composite carries closure if and only if both legs carry it and the intermediate contract matches. The side conditions are named rather than assumed away: the unit law needs the contract test to be reflexive, the equational forms of the laws need extensionality, and obligation-totality composition needs containment of demanded obligations rather than a registered bridge. One consequence is negative and is kept: this paper's composition rule is provably **incomplete against its own obligation semantics**, refusing composites whose contracts demand exactly the same obligations when no bridge is registered. It is sound and fail-closed, not exact.

The bounded model is then recovered as an instance rather than repeated as separate support. The five donor families are interpreted as a **transformation family**: each is a transformation with its own source and target obligation contracts, and the intermediate-contract test becomes a function of the two transformations and of the registered bridge relation rather than a value supplied by the caller. Under three frame conditions, twelve further theorems are discharged at arbitrary width, and this paper's published 25 composition successes and 25 bridge-mismatch countermodels are recomputed by running the committed implementation with the hand-off computed rather than typed.

What that recovery exposes is reported in full below, because it is a limit on the paper's own evidence and not on the theorems.

The composition calculus

Let `Trans`, `Contract`, `Coord` and `Obl` be sorts, `Native : Trans -> Bool` the donor's own preservation/refinement/round-trip verdict, `Holds : Trans x Coord -> Bool` the per-coordinate closure status, `Src`, `Tgt : Trans -> Contract` the obligation contracts a transformation consumes and emits, and `Bridge : Contract x Contract -> Bool` the registered bridge relation.

The closure lift. `Carries(t) <-> Native(t) AND forall c. Holds(t, c)`. This is V3's `ClosureCarries(T, o)` with the five-tuple replaced by a quantifier, so nothing below depends on there being five coordinates.

The intermediate-contract test. `Match(a, b) := a = b OR Bridge(a, b)`, verbatim from V3.5's "exactly the same contract, or a registered bridge".

Coordinate transport (the one substantive axiom). A composite holds a closure coordinate exactly when both legs hold it, and the distinguished coordinate `obligations_total` additionally requires `Match(Tgt t, Src u)`.

Theorem V4.1 — intermediate-contract composition

`Carries(Comp(t,u)) <-> Carries(t) AND Carries(u) AND Match(Tgt t, Src u)`, for every pair of transformations. This is V3.5, and it is *derived* from coordinate transport rather than assumed; a calculus that took it as an axiom would be assuming its own conclusion.

Theorem V4.2 — unit and associativity

`Ident(a)` carries closure, is a two-sided unit, and the two bracketings of a three-transformation chain are observationally equal — with nothing assumed of the bridge relation: no reflexivity, no symmetry, no transitivity. Under extensionality both laws hold as equations. The unit law needs the *test* to be reflexive, which it is because `Match` has an equality disjunct; with the raw registered relation in its place there is a countermodel.

Theorem V4.3 — the rule is sound and incomplete

Under an obligation semantics defined without reference to any composition rule, `Total(t) <-> forall o. Demands(Tgt t, o) -> Demands(Src t, o) OR Discharged(t,o) OR Fresh(t,o)`, matching intermediate contracts **suffice** for totality to compose but are **not necessary**. The exact condition is weaker: containment of the second leg's source demands in the first leg's target demands. This paper's rule therefore refuses composites that are in fact obligation-total. This is a real gap in the paper's own rule and is reported as one.

19. The donor stack as a transformation family

V3's composition support is a `5 x 5` loop over donor families in which, in the committed checker, neither donor appears on the right-hand side of anything and both legs are the same loop-invariant expression. Instrumented, that loop evaluates `compose` at **2 of its 8 possible argument triples**. Under the interpretation each family becomes a transformation with its own hand-off contracts, and the third argument is computed from the pair rather than typed.

Three frame conditions are stated as axioms rather than folded into an encoding:

1. **Hand-offs are never contract identities** — no family's target contract is any family's source contract, its own included.
2. **Distinct families have distinct endpoints.**
3. **The stack is inhabited.**

Theorem V4.4 — the composition rows, generalised

A bridged hand-off between two carrying donors composes; an unbridged one refuses; a donor composed with itself still needs a bridge registered from its own target contract back to its own source contract; and a chain of three donors carries exactly when all three legs carry and both interior hand-offs are bridged. For a stack of any size.

Theorem V4.5 — the rows the enumeration does not contain

If either leg fails to carry, the composite does not carry, whatever the other leg is and whatever the registry says. These are the six argument triples the published loop never evaluates.

Theorem V4.6 — the donor axis indexes something

Distinct ordered donor pairs present distinct hand-offs, and two pairs whose hand-offs the registry decides differently have composites that carry differently. On an inhabited carrying stack, both the successes and the refusals have witnesses rather than holding vacuously.

Two conditions that looked like frame conditions — that no identity is a donor, and that a composite's hand-off is separated too — were written as axioms first, reported inert, and are now discharged as theorems from the remaining three. Their inertness is re-measured rather than remembered: both are added back to the axiom set on every run and the theorems that becomes newly provable are reported, and the answer has to stay none.

20. What the published composition counts do and do not carry

This section exists because the numbers in V3's support table are larger than the facts behind them, and the difference is measurable.

25 bridge-mismatch countermodels is exactly one frame condition. Once the hand-off is computed rather than typed, a refused composite is one whose hand-off does not match, so 25 refusals under an empty registry *is* the statement that no family's target contract is any family's source contract. Read as a check on the interpretation this is good news: the condition is this paper's own assertion and not a modelling convenience. Read as a count it is one fact. Assignments that violate it return 0, 20 and 21 instead of 25.

25 composition successes discriminates nothing about the interpretation. Every contract assignment tried — including ones that collapse the whole stack to a single contract pair — returns 25 under a registry that bridges every hand-off, because a full registry satisfies the match test whatever the contracts are.

The counts cannot see the second frame condition at all. Reading every family as consuming one shared input contract and emitting one shared output contract reproduces both published numbers exactly, and so does reading them as emitting one shared output contract. Those are the readings in which the `5 x 5` loop visits **1** and **5** distinct hand-offs rather than 25 — precisely the multiplier the interpretation is supposed to remove — and the published pair is blind to both. What separates them is Theorem V4.6 and the directly measured hand-off count, not the counts. A registry that bridged some hand-offs and not others would have separated them empirically; P7 shipped no such registry.

The published counts still reach two argument triples, and this is now a fact about the registries. Reproducing 25 successes requires every hand-off bridged and reproducing 25 countermodels requires none bridged, both over an all-carrying stack. The 25 rows are 25 distinct hand-offs rather than 25 copies of one argument; they remain 25 agreeing verdicts. The other six triples are exercised separately, over 204,800 evaluations of the committed functions against the proved rule. Both verdicts occur, but the corpus is heavily skewed — only 25 of those rows compose successfully, one per ordered donor pair, because a leg carries at exactly one of the 64 (native verdict, closure vector) rows it is given — and the per-triple counts are published beside the total rather than summarised. That enumeration is not part of this paper's published result and does not become part of it by being run.

Exhaustive bounded support (unchanged, re-read)

The enumeration of V3 is unchanged on disk and its digest is unchanged:

- states: **320**; donor-conservativity violations: **0**; ideal-product mismatches: **0**;
- minimal one-coordinate closure separations: **25**;
- donor-product nonclosure countermodels: **31**;
- exact full closure-refinement successes: **155**;
- proper-subset closure-refinement failures: **1,055**;
- heterogeneous composition successes under exact bridge: **25**;
- bridge-mismatch composition countermodels: **25**;
- canonical rows SHA-256: `25f40385714adb15bca298a8cfd2b7fe2b28c96bfe462f6b60583be8f735b95f`.

What V4 changes is the reading. The 155 and 1,055 are a five-way donor multiplication of 31 and 211: the donor loop enters neither the closure vector nor the repair set. Neither does anything else under it — 320 states is the 64-point (native verdict, closure vector) space enumerated once per family, the 25 minimal separations are 5 counted five times, and the 25 successes and 25 bridge countermodels are one of each counted once per ordered donor pair. Only the 31 nonclosure countermodels are 31 distinct facts, and the `donor\_axis` block of the result artifact carries the same table. The composition pair is discussed in §20. The counts remain correct and remain reproducible; they are support for a bounded instance, and the general claims are now carried by §§18–19 instead.

Wider P7 claim (V4 wording)

> Scientific navigation can reuse mature planning/refinement, counterexample-guided reopening, representation-migration, replanning and terminal-commitment machinery while carrying task-global closure as an explicit obligation object; closure-transport defects can be selectively refined; and heterogeneous navigation transforms compose scientifically only when their intermediate closure contracts are correctly bridged — which holds for chains of any length over donor stacks of any size, and is sound but not complete against this paper's own obligation semantics.

Non-synthetic evidence in three change classes

The support enumerated in §5 is bounded and synthetic. Three executed studies carry the same separation into real data, one per change class: representation, responsibility/ontology, and objective/obligation. Each was frozen before execution and reproduced by a second implementation that shares no facts or gold code with the first.

Representation change. The public RO-Crate `1.2->1.3` transition, in which four Bioschemas workflow terms changed canonical URI bindings while ordinary value-level JSON usage can appear unchanged. Across 14 frozen standard-transition cases, witness-aware transport is exact (**1.0**), with **0** false closures, **0** unnecessary reopens and **4** correct `CANNOT\_CHECK` dispositions. Value-only transport reaches **0.428571** with **8** false closures; always-reopen reaches **0.285714** with **6** unnecessary reopens.

Responsibility/ontology change. The 178-example UCI Wine recognition data, where the coarse responsibility `class0\_vs\_other` makes the reverse coarse-to-fine map non-injective: coarse value `0` merges fine classes `1` and `2`. Across 712 protected transport rows, witness-aware transport is exact (**1.0**) with **238** correct `CANNOT\_CHECK`; value-only reaches **0.665730** and converts those same 238 into false closures; always-reopen reaches **0.0** with **474** unnecessary reopens. The **119** ambiguous coarse-0 examples are the cases where locally well-formed representation changes require different dispositions depending on retained support history.

Objective/obligation change. The 569-example Wisconsin Diagnostic Breast Cancer data, under a frozen `StratifiedKFold(n\_splits=5,shuffle=True,random\_state=20261217)` split. The obligation changes from aggregate accuracy to malignant recall, and the two obligations come apart in both directions: fold 0 satisfies the old accuracy obligation (`0.965>=0.95`) but fails the new recall obligation (`0.907<0.95`), while fold 1 fails the old (`0.947<0.95`) and satisfies the new (`0.953>=0.95`). All five accuracy-only cells are `CANNOT\_CHECK` under the changed obligation, because malignant recall is not inferable from aggregate accuracy; value-only produces **5** false closures and always-reopen **4** unnecessary reopens.

Witness-aware transport is **1.0** in all three classes, and the two degenerate policies separate in every one: value preservation alone closes transitions it has not earned, and always reopening pays for transitions that were already witnessed.

What this does not establish. These are three executed change classes, not a universal claim. The result does not establish scientific-regime transport across arbitrary world-model, objective or research-agent changes, and it does not establish independent scientific authority: the agreement is between two implementations inside the same workflow, not an external adjudication. The breast-cancer study is a transport experiment over a frozen split. No clinical use is claimed, and no classifier or medical decision rule is offered.

Limits

The registered closure coordinates remain a bounded formal instance and are not claimed universally minimal. The composition rule is fail-closed and provably incomplete: it refuses composites that are obligation-total but unbridged. No naturalistic multi-stage pipeline corpus exists, so nothing here is an empirical or deployed-agent claim, and the false-closure/recomputation trade-off is not measured. No formal or empirical reproduction outside the producing lane has been arranged: the calculus, the interpretation and the tests were written in the same lane, and independent reproduction remains open. This paper's other formal core — the 64-state support-transport theorem — is a different model and is untouched here.

Current science terminal:
`P7_CLOSURE_CARRYING_NAVIGATION_SUPPORTED__BOUNDED_FORMAL_DONOR_STACK__IDEAL_PRODUCT_EQUIVALENT`.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the author used OpenAI ChatGPT and Codex for drafting, editing, source checking, adversarial review, and submission-package preparation. The author reviewed the output and takes full responsibility for the publication's content.

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